

Hybrid Oscillator-Qubit Linear Combination of Hamiltonian Simulation

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Abstract

We propose an advanced method for quantum simulation based on the linear combination of Hamiltonian simulation (LCHS) within a hybrid oscillator-qubit architecture. This approach capitalizes on the inherent advantages of continuous (oscillator) and discrete (qubit) variables, eliminating the need for discretization and the associated truncation errors commonly found in discrete variable LCHS. This work marks an important step forward in overcoming the limitations of traditional simulation methods and advancing the capabilities of quantum computing.

1 Introduction

2 Motivation

Linear combination of Hamiltonian simulation (LCHS) [1, 2] provides an optimal state preparation cost and a near-optimal scaling in matrix queries on all parameters for solving linear ordinary differential equations (ODEs) in the form of

$$\frac{du(t)}{dt} = -A(t)u(t) + b(t), \quad u(0) = u_0 \quad (1)$$

where $t \in [0, T]$, $A(t) \in \mathbb{C}^{N \times N}$ and $u(t), b(t) \in \mathbb{C}^N$. The solution of the ODE (1) has the explicit form

$$u(T) = \mathcal{T}e^{-\int_0^T A(s)ds}u_0 + \int_0^T \mathcal{T}e^{-\int_s^T A(s')ds'}b(s)ds \quad (2)$$

where $\mathcal{T}$ is the time-ordering operator. To start with, the coefficient matrix $A(t)$ is first decomposed into

$$A(t) = L(t) + iH(t)$$

via Cartesian decomposition

$$L(t) = \frac{A(t) + A^\dagger(t)}{2}, H(t) = \frac{A(t) - A^\dagger(t)}{2i}.$$

It is assumed that $L(t) \succeq 0$. Otherwise, we can shift the eigen spectrum of $u(t)$ by substituting $u(t) = e^{ct}v(t)$ and use $v(t)$ for the ODE system so that $L(t) + cI \succeq 0$, where $-c$ is the minimum of the smallest eigenvalues of $L(t)$. LCHS provides that

$$\mathcal{T}e^{-\int_0^t A(s)ds} = \int_{\mathbb{R}} g(k) \left[\mathcal{T}e^{-i\int_0^t (kL(s) + H(s))ds} \right] dk \quad (3)$$

$$\int_0^t \mathcal{T}e^{-\int_s^t A(s')ds'}b(s)ds = \int_0^t \int_{\mathbb{R}} g(k) \left[\mathcal{T}e^{-i\int_s^t (kL(s') + H(s'))ds'} \right] b(s)dkds \quad (4)$$

where $g(k) := \frac{f(k)}{1-ik}$ for some kernel function $f(k)$ and $t \in [0, T]$. The choice of $f(k)$ that leads to near-optimal complexity dependency on all parameters is

$$f(k) = \frac{e^{2\beta}}{2\pi e^{(1+ik)^\beta}},$$

where $\beta \in (0, 1)$ is a user-defined constant [2]. Note that the terms in brackets in Eq. (3) and (4) are unitary. Thus, LCHS further truncates and discretizes the integrals in Eq. (3) and (4) into a sum of unitaries, employing

state preparation circuits for $|u_0\rangle := \frac{u_0}{\|u_0\|}$ and $|b(s)\rangle := \frac{b(s)}{\|b(s)\|}$ and the linear combination of unitaries (LCU) circuits to prepare the solution state $|u(T)\rangle := \frac{u(T)}{\|u(T)\|}$ [1, 2].

Combining with the finite difference method and linearization methods, for example, Carleman linearization [3], various non-linear partial differential equations can also be converted into a linear ODE system that is solvable by LCHS. However, the number of terms in the discretized LCHS integral, denoted by M , brings a huge cost on quantum resources in the circuit implementation level for either near-term or early fault-tolerant quantum machines. For the complementary solution, M has complexity

$$M \in \mathcal{O} \left(T \max_t \|L(t)\| \left(\log \frac{1}{\epsilon} \right)^{1+1/\beta} \right)$$

where ϵ is the total error incurred by LCHS [2]. The corresponding LCU circuit requires $\lceil \log_2(M) \rceil$ ancillary qubits, and more problematically, $\lceil \log_2(M) \rceil$ -controlled multiple-qubit unitary operators for Hamiltonian simulations. In this case, for a time-independent ODE, the value of M can be in the 10^6 magnitude when ϵ is around 10^{-5} , $\beta = 0.8$, $T = 1000$, and $\|L\| = 1$. More exact scalings of LCHS are provided in [4].

Inspired by the continuous LCU in [5], a hybrid CV-DV LCHS is a promising direction to bring down the resource cost and, ideally, remove truncation and discretization errors in the integrals calculation. Specifically, continuous LCU can use $\mathcal{O}(1)$ number of ancillary oscillators, along with CV state preparation and CV measurements, replace $\lceil \log_2(M) \rceil$ ancillary qubits in the DV LCU circuit, while the unitaries for Hamiltonian evolution remain in qubit space. Continuous LCU circuit can compute the integrals in Eq. (3) and (4) without truncation and discretization on integrals, whereas truncations errors may still persist in CV state preparation and hybrid Hamiltonian simulation.

3 Methodology

Let us consider a time-independent homogeneous ODE system, where $A(t) := A$ and $b(t) := 0$ for all t . From Eq. (2) and (3), the complementary solution becomes

$$u(t) = \left[\int_{\mathbb{R}} g(k) e^{-it(kL+H)} dk \right] u_0 \quad (5)$$

for $t \in [0, T]$. Using a single oscillator for the ancillary in continuous LCU, the qubits that synthesize unitary $e^{-it(kL+H)}$ need to couple with this ancillary oscillator, leading to the joint evolution $e^{-it(L \otimes \hat{k} + H \otimes I)}$, where $\hat{k}$ is one of the canonical conjugation operators ($\hat{x}$ and $\hat{p}$) and I is the identity operator. That is to say, the qumode remains untouched when H acts on the qubit system.

Before forming the circuit for the joint evolution, an appropriate function $g(k)$ has to be obtained. To do this, we need to prepare and post-select two appropriate CV states.

Let

$$|\Psi\rangle = |\psi\rangle_{\text{osc}} |u_0\rangle_q$$

be the preparation state, where $|\psi\rangle_{\text{osc}}$ is the initial oscillator state and $|u_0\rangle$ is the initial qubit state.

Let

$$\langle\Phi| = \langle\phi|_{\text{osc}} \otimes \mathbf{I}_q$$

be the post-selection projector, where $\langle\phi|_{\text{osc}}$ acts only on the oscillator and the qubit remains untouched.

If we choose the position operator $\hat{x}$ for k , then according to ref. [5], the qumode sandwich gives

$$\langle\Phi| e^{-it(\hat{x} \otimes L + \mathbf{I}_{\text{osc}} \otimes H)} |\Psi\rangle = \left[\int_{\mathbb{R}} dx \phi^*(x) \psi(x) e^{-it(xL+H)} \right] |u_0\rangle_q.$$

Comparing to the Eq. (5), this gives

$$g(x) = \phi^*(x) \psi(x)$$

Namely, we need to engineer the preparation and post-selection states to obtain an appropriate $g(x)$ that satisfies LCHS's requirements.

We choose the post-selection state $|\phi\rangle$ to be a squeezed vacuum, whose position-space wavefunction is

$$\phi_r(x) := S(r, 0) \phi_0(x) = \left(\frac{2}{e^{2r}\pi} \right)^{1/4} e^{-x^2/e^{2r}} = \left(\frac{2}{\sigma^2\pi} \right)^{1/4} e^{-x^2/\sigma^2}$$

with squeezing parameter r and wavefunction standard deviation $\sigma = e^r$, and this gives the corresponding initial state to be prepared

$$\psi_r(x) := \left(\frac{\sigma^2\pi}{2}\right)^{1/4} g(x)e^{x^2/\sigma^2} = \mathcal{N}g(x)e^{x^2/\sigma^2}. \quad (6)$$

This leads to our initial formalism of a CV/DV LCHS theorem:

Theorem 1 (CV/DV LCHS). *Suppose $A = L + iH$ and $L \succeq 0$, then $u(t)$ can be expressed as the following:*

$$|u(t)\rangle = e^{-At} |u(0)\rangle \quad (7)$$

$$= (\langle\phi|_{\text{osc}} \otimes \mathbf{I}_q) e^{-it(\hat{x} \otimes L + \mathbf{I}_{\text{osc}} \otimes H)} (|\psi\rangle_{\text{osc}} \otimes |u_0\rangle_q) \quad (8)$$

$$= \left[\int_{\mathbb{R}} dx \phi^*(x) \psi(x) e^{-it(xL+H)} \right] |u_0\rangle_q. \quad (9)$$

where $|\psi\rangle_{\text{osc}}$ and $|\phi\rangle_{\text{osc}}$ are squeezed Fock states.

At this point, the missing piece for the whole hybrid CV-DV LCHS circuit is the oracle syntheses for matrices L and H , whose basis-gate implementations are highly dependent on the structures of both matrices and are very problem-specific. Many previous works in the DV regime tackle this problem by decomposing coefficient matrices into linear combinations of Pauli matrices, for example, hyperbolic partial differential equations [6] and advection-diffusion equation [7]. Borrowing their designs and following the circuit synthesis in [5], the evolution of decomposed Hamiltonians coupled with oscillators can be constructed in CV-DV circuits.

3.1 State Preparation

We now consider two special cases of $\psi_r(x)$ preparation and their implications.

- $\mathbf{r} = \mathbf{0}$: This leads to the post-selection state being the vacuum state $\phi_0(x)$. However, our preparation state becomes

$$\psi_0(x) := \mathcal{N}g(x)e^{x^2} \quad (10)$$

which is not a physically realizable state, as $\psi_0(x) \notin L^2(\mathbb{R})$.

This gives us an additional condition imposed on the $g(x)$ for CV-DV LCHS:

$$\exists r > 0 \text{ s.t. } \int_{\mathbb{R}} dx |g(x)|^2 e^{\frac{2x^2}{e^{2r}}} < \infty$$

- $\mathbf{r} \rightarrow \infty$: This leads to $\phi_r(x)$ being the infinitely squeezed vacuum state. Then our preparation state is simply the kernel itself:

$$\psi_\infty(x) := \lim_{r \rightarrow \infty} \psi_r(x) = \mathcal{N}g(x) \quad (11)$$

The initial state $|\psi_r\rangle$ can be approximated via a finite squeezed Fock expansion

$$\psi_r(x) = \mathcal{N}g(x)e^{x^2/e^{2r}} \approx \sum_{n=0}^{N_{\max}} C_n \langle x|S(r,0)|n\rangle = \psi'(x)$$

where coefficients $\{C_n\}$ are overlaps

$$C_n = \langle n|S^\dagger(r',0)|\psi_r\rangle = \int_{\mathbb{R}} dx \phi_n^*(x/\sigma') e^{-x^2/\sigma^2} \psi_r(x)$$

This expression simplifies to

$$C_n = \sqrt{\frac{\sigma}{\sigma'}} \frac{1}{\sqrt{2^n n!}} \int_{\mathbb{R}} dx H_n(x/\sigma') g(x) e^{-\gamma x^2} = \int_{\mathbb{R}} \frac{dx H_n(x/\sigma') e^{-\gamma x^2}}{2\pi e^{-2\beta} e^{(1+ix)^\beta} (1-ix)} \quad (12)$$

where $\gamma = \frac{1}{\sigma'^2} - \frac{1}{\sigma^2} = \frac{1}{e^{2r'}} - \frac{1}{e^{2r}}$, $r' < r$.

3.2 Analytical Bounds on the Expansion Coefficients C_n

To characterize the behavior of the projection coefficients C_n for $\beta \in [0, 1]$, we establish rigorous analytical bounds by evaluating the integral at the extremal values of the parameter. The coefficients are defined by the projection of the signal $g(x)$ onto the Hermite basis:

$$C_n(\beta) = \mathcal{K}_n \int_{-\infty}^{\infty} \frac{H_n(x/\sigma') e^{-\gamma x^2}}{2\pi e^{-2\beta} e^{(1+ix)^\beta} (1-ix)} dx \quad (13)$$

where $\mathcal{K}_n = \sqrt{\frac{\sigma}{\sigma'}} \frac{1}{\sqrt{2^n n!}}$ and $\gamma = \sigma'^{-2} - \sigma^{-2}$. We denote the lower and upper bounds as $C_n^{(0)}$ and $C_n^{(1)}$, corresponding to the limits $\beta = 0$ and $\beta = 1$, respectively.

We employ an explicit polynomial decomposition of the Hermite-Lorentzian kernel. By defining a residual quotient polynomial $Q_{n-1}(x) = [H_n(x/\sigma') - H_n(-i/\sigma')]/(x+i)$, we isolate the singular contribution of the complex pole at $x = -i$.

At the limit $\beta = 0$, the denominator simplifies to a standard complex Lorentzian. The exact closed-form expression is:

$$C_n^{(0)} = \mathcal{K}_n \left[\frac{1}{2} H_n \left(-\frac{i}{\sigma'} \right) e^{\gamma} \operatorname{erfc}(\sqrt{\gamma}) + \frac{i}{2\pi} \int_{-\infty}^{\infty} Q_{n-1}(x) e^{-\gamma x^2} dx \right] \quad (14)$$

For $\beta = 1$, the kernel incorporates a phase shift e^{-ix} and a scaling factor e^1 . Completing the square in the exponent for the shifted Gaussian $(-\gamma x^2 - ix)$ leads to:

$$C_n^{(1)} = \mathcal{K}_n \left[\frac{1}{2} H_n \left(-\frac{i}{\sigma'} \right) e^{\gamma} \operatorname{erfc} \left(\sqrt{\gamma} - \frac{1}{2\sqrt{\gamma}} \right) + \frac{ie}{2\pi} \int_{-\infty}^{\infty} Q_{n-1}(x) e^{-\gamma x^2 - ix} dx \right] \quad (15)$$

The transition from $\beta = 0$ to $\beta = 1$ effectively shifts the argument of the error function by $\Delta = (2\sqrt{\gamma})^{-1}$.

3.2.1 Numerical Validation

The analytical results were compared against direct numerical quadrature using a double-exponential integration scheme with a working precision of 30 decimal places. For the parameter set $\sigma' = 1.2$ and $\sigma = 2.0$ ($\gamma = 4/9$), the agreement is presented in the tables below.

Table 1: Comparison of Numerical and Analytical Coefficients for the Lower Bound ($\beta = 0, \gamma = 4/9$)

n	Numerical Result	Analytical (Explicit)
0	0.3481065885	0.3481065885
1	0.2335427879 <i>i</i>	0.2335427879 <i>i</i>
2	-0.0515295394	-0.0515295394
3	-0.0077856810 <i>i</i>	-0.0077856810 <i>i</i>

Table 2: Comparison of Numerical and Analytical Coefficients for the Upper Bound ($\beta = 1, \gamma = 4/9$)

n	Numerical Result	Analytical (Explicit)
0	1.1011786413	1.1011786413
1	-0.3006294949 <i>i</i>	-0.3006294949 <i>i</i>
2	-0.0943736080	-0.0943736080
3	-0.1769037943 <i>i</i>	-0.1769037943 <i>i</i>

As demonstrated in Tables 1 and 2, the explicit decomposition method maintains a relative error below 10^{-25} . This confirms the stability of the analytical framework and establishes robust expansion bounds across the entire parameter space of β .

3.3 Error bound estimation for state preparation

Ideally we want to realize $\psi_r(x) = g(x)e^{x^2/e^{2r}}$ as close as possible using Gaussian (Fock) states. Previously, authors from [8] showed that sub-optimal kernel function $f(k) = \frac{1}{\pi(1+ik)}$ could be realized with a squeezed vacuum state with maximum fidelity of 98.6% with a squeezing parameter of 0.925. However, we show that the near-optimal kernel function can be realized instead with a superposition of squeezed Fock states for arbitrary closeness, where the approximation error is bounded by the Fock cutoff level (or the oscillator energy) and the squeezing parameter:

Theorem 2. Let $\psi_r(k)$ denote the initial preparation state, and let $\psi'(k)$ be an ϵ_s -approximation to $\psi_r(k)$. For any $\epsilon_s > 0$, there exists a Fock cutoff level $N \in \mathbb{N}$ and squeezing parameter $r > 0$ such that

$$\|\psi'(k) - \psi_r(k)\| < \epsilon_s.$$

where the required resources scale asymptotically as

$$N = O\left(\log^2 \frac{1}{\epsilon_s}\right), \quad r = \Omega\left(\log \frac{1}{\epsilon_s}\right).$$

Proof. Define $\tau(N) = \|\psi'(k) - g(k)\|$ and $\delta(r) = \|g(k) - g(k)e^{k^2/e^{2r}}\|$. Then we can write

$$\epsilon_s(r, N) \leq \tau(N) + \delta(r).$$

Define the truncated Hilbert space $\mathbb{H}_N = \text{span}\{\psi_n\}_{n=0}^N$, and let $\Pi_N^F : L^2(\mathbb{R}) \rightarrow \mathbb{H}_N$ be the orthogonal projection. Then the projection of g is

$$(\Pi_N^F g)(k) = \sum_{n=0}^N C_n \psi_n(k) = \psi'(k).$$

Thus,

$$\|g(k) - \psi'(k)\| \leq \|g - \Pi_N^F g\|.$$

g is analytic in the strip $\mathcal{S}_\rho := \{z \in \mathbb{C} : \Im(z) \in [-\rho, \rho]\}$ for $\rho = 1 - \epsilon$ with $\epsilon \rightarrow 0$; By the convergence analysis of Hermite approximations [9, Thm. 3.8], under the stated analyticity¹, the error satisfies

$$\tau = \|g - \Pi_N^F g\| \leq \mathcal{K}e^{-\rho\sqrt{2N}}.$$

Now we write $\delta(r) = \|g(k) - g(k)(1 - k^2/e^{2r} + k^2O(e^{-2r}))\| \leq \mathcal{M}e^{-2r}$. Then,

$$\begin{aligned} \epsilon_s &\leq \mathcal{M}e^{-2r} + \mathcal{K}e^{-\rho\sqrt{2N}} \\ \implies N &\leq \frac{1}{2\rho^2} \log^2\left(\frac{\mathcal{K}}{\epsilon_s - \mathcal{M}e^{-2r}}\right) \end{aligned}$$

Since the argument must be positive, $\epsilon_s - \mathcal{M}e^{-2r} \geq 0 \implies r > \log(\mathcal{M}/\epsilon_s)$. Under this bound,

$$N \leq \log^2(\mathcal{K}/\epsilon_s)/(2\rho^2).$$

□

3.4 Hamiltonian evolution approximation

Since the hybrid Hamiltonian $\hat{x} \otimes \hat{L} + I \otimes \hat{H}$ is not directly realizable as a native interaction, its time evolution is synthesized via a Trotter-Suzuki decomposition into a sequence of implementable operations.

Let $\delta t = t/n$, where t is the total simulation time and n is the number of Trotter steps, and define

$$A = \hat{x}_N \otimes \hat{L}, \quad B = I \otimes \hat{H}.$$

where the operator $\hat{x}_N$ is understood on a suitably truncated or energy-constrained CV subspace. The first-order Lie-Trotter formula yields $U(t) = e^{-it(A+B)} \approx (e^{-i\delta t A} e^{-i\delta t B})^n$, with error scaling as $\epsilon_t = \mathcal{O}\left(\frac{t^2}{n} \|[A, B]\|\right)$.

Since $[A, B] = \hat{x}_N \otimes [\hat{L}, \hat{H}]$, we obtain the bound $\|[A, B]\| \leq \|\hat{x}_N\| \|\hat{L}, \hat{H}\|$. Consequently,

$$n = \mathcal{O}\left(\frac{t^2}{\epsilon_t} \|\hat{x}_N\| \|\hat{L}, \hat{H}\|\right). \quad (16)$$

Suppose that the Hamiltonians admit Pauli decompositions of the form

$$\hat{L} = \sum_{i=1}^{N_L} \alpha_i P_i, \quad \hat{H} = \sum_{j=1}^{N_H} \beta_j P'_j, \quad (17)$$

¹The strict version of the theorem states that g must satisfy some growth conditions as well. However, the authors imply in the same work that the growth conditions can be relaxed [9, pp. 15]. Here we assume the relaxed version of the theorem.

where P_i and P'_j are Pauli strings. Each exponential can then be further decomposed using a first-order product formula as

$$e^{-i\delta t A} \approx \prod_{i=1}^{N_L} e^{-i\delta t \alpha_i (\hat{x}_N \otimes P_i)}, \quad e^{-i\delta t B} \approx \prod_{j=1}^{N_H} e^{-i\delta t \beta_j (I \otimes P'_j)}. \quad (18)$$

Combining these decompositions yields the approximation

$$U(t) \approx \left(\prod_{i=1}^{N_L} e^{-i\delta t \alpha_i (\hat{x}_N \otimes P_i)} \prod_{j=1}^{N_H} e^{-i\delta t \beta_j (I \otimes P'_j)} \right)^n, \quad (19)$$

up to additional error arising from the non-commutativity of the Pauli terms.

Each Pauli rotation $e^{-i\theta P}$ is implemented using a standard basis-change and CNOT-ladder construction, requiring $2(w(P) - 1)$ CNOT gates, where $w(P)$ denotes the Pauli weight. Consequently, the total number of CNOT gates scales as

$$N_{\text{CNOT}} = \left(2n \sum_{k=1}^{N_L + N_H} (w(P_k) - 1) \right), \quad (20)$$

where $\{P_k\}$ denotes the collection of Pauli strings appearing in the decomposition of $\hat{L}$ and $\hat{H}$.

In addition, each Trotter step contains N_L hybrid CV-DV interactions of the form $e^{-i\delta t \alpha_i (\hat{x}_N \otimes P_i)}$, yielding a total of

$$N_{\text{hyb}} = n N_L$$

hybrid gates.

Defining the Pauli 1-norms

$$\|\hat{L}\|_1 := \sum_i |\alpha_i|, \quad \|\hat{H}\|_1 := \sum_j |\beta_j|, \quad (21)$$

and using the bound $\|[\hat{L}, \hat{H}]\| \leq 2\|\hat{L}\|_1 \|\hat{H}\|_1$, the first-order Trotter error implies

$$n = O\left(\frac{t^2}{\epsilon_t} \|\hat{x}_N\| \|\hat{L}\|_1 \|\hat{H}\|_1\right). \quad (22)$$

Therefore, the overall CNOT complexity scales as

$$N_{\text{CNOT}} = O\left(\frac{t^2}{\epsilon_t} \|\hat{x}_N\| \|\hat{L}\|_1 \|\hat{H}\|_1 \sum_k w(P_k)\right). \quad (23)$$

Higher-order product-formula simulation. For an even order $p = 2k$, the p -th order Suzuki formula yields

$$U(t) = e^{-it(\hat{x}_N \otimes \hat{L} + I \otimes \hat{H})} \approx (S_p(\delta t))^n, \quad (24)$$

with error scaling

$$\epsilon_t = \mathcal{O}\left(\frac{t^{p+1}}{n^p} \|\hat{x}_N\| \Gamma_p(\hat{L}, \hat{H})\right), \quad (25)$$

For operators X and Y , we define the r -fold nested commutator recursively by

$$[X^{(0)}, Y] := Y, \quad [X^{(r)}, Y] := [X, [X^{(r-1)}, Y]]. \quad (26)$$

Then we can define $\Gamma_p(\hat{L}, \hat{H})$ in terms of nested commutators as

$$\Gamma_p(\hat{L}, \hat{H}) = \max_{\substack{r+s=p \\ r,s \geq 0}} \left\{ \|[\hat{L}^{(r)}, [\hat{H}^{(s)}, \hat{L}]]\|, \|[\hat{L}^{(r)}, [\hat{H}^{(s)}, \hat{H}]]\| \right\}. \quad (27)$$

For practical purposes, this quantity admits the upper bound

$$\Gamma_p(\hat{L}, \hat{H}) \leq C_p (\|\hat{L}\| + \|\hat{H}\|)^{p-1} \|\hat{L}, \hat{H}\|. \quad (28)$$

Consequently,

$$n = \mathcal{O}\left(t^{1+1/p} \left(\frac{\|\hat{x}_N\| \Gamma_p(\hat{L}, \hat{H})}{\epsilon_t}\right)^{1/p}\right). \quad (29)$$

Moreover,

$$[\hat{x}_N \otimes \hat{L}, I \otimes \hat{H}] = \hat{x}_N \otimes [\hat{L}, \hat{H}], \quad (30)$$

which implies the bound

$$\|[\hat{x}_N \otimes \hat{L}, I \otimes \hat{H}]\| \leq \|\hat{x}_N\| \|[\hat{L}, \hat{H}]\|, \quad (31)$$

where $\|\hat{x}_N\|$ is evaluated on a suitably truncated or energy-constrained CV subspace.

Gate counts. Let m_p denote the number of exponentials appearing in a single application of the p -th order Suzuki formula $S_p(\delta t)$. For the standard recursive construction, one has

$$m_2 = 3, \quad m_{2k} = 5 m_{2k-2}, \quad \Rightarrow \quad m_p = 3 \cdot 5^{(p/2)-1}. \quad (32)$$

The total number of CNOT gates scales as

$$N_{\text{CNOT}} = \left(2nm_p \sum_{k=1}^{N_L+N_H} (w(P_k) - 1) \right), \quad (33)$$

where $\{P_k\}$ denotes the collection of Pauli strings arising from the Pauli decompositions of $\hat{L}$ and $\hat{H}$.

Moreover, each exponential of the form $e^{-i\theta(\hat{x} \otimes P_i)}$ contributes a hybrid CV-DV interaction. Let c_p denote the number of such hybrid exponentials within a single application of $S_p(\delta t)$. For the standard Suzuki construction, one has $c_p = \Theta(m_p)$, and therefore the total number of hybrid gates scales as

$$N_{\text{hyb}} = O(nc_p N_L). \quad (34)$$

3.5 Oracle Synthesis

3.5.1 Toy Example: Damped Harmonic Oscillator

Consider a damped harmonic oscillator with position $x(t)$, damping rate $\gamma \geq 0$, and angular frequency $\omega > 0$. Its equation of motion is

$$\frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + \omega^2 x = 0. \quad (35)$$

Introducing the velocity variable $v = \dot{x}$, this second-order equation can be written as a first-order linear system

$$\frac{d}{dt} \begin{bmatrix} x \\ v \end{bmatrix} = - \begin{bmatrix} \gamma/2 & -\omega \\ \omega & \gamma/2 \end{bmatrix} \begin{bmatrix} x \\ v \end{bmatrix}. \quad (36)$$

We denote the coefficient matrix by A .

To cast the dynamics into the LCHS framework, we perform the Cartesian decomposition of A ,

$$L = \frac{A + A^\dagger}{2} = \frac{\gamma}{2} I, \quad (37)$$

$$H = \frac{A - A^\dagger}{2i} = -\omega Y, \quad (38)$$

where I is the identity matrix and Y is the Pauli- Y matrix.

The corresponding unitary operators appearing in the LCHS circuit construction are therefore

$$\exp(-iT k L) = \exp\left(-iI \frac{\gamma T k}{2}\right), \quad (39)$$

$$\exp(-iT H) = \exp(i\omega T Y). \quad (40)$$

Thus, the dissipative contribution is captured by a uniform decay factor, while the Hamiltonian part corresponds to a planar rotation generated by Y . The resulting LCHS simulation can be implemented using single-qubit R_Y rotations together with non-unitary damping steps, with standard Trotterization over discrete time steps.

To-Do, 2/26/26

1. compile with qcvdv and show gate count
2. compare qumode energy vs different qubit architectures energy (prioritize superconducting)

3.5.2 Example: Discretized One-dimensional Heat Equation

Consider one-dimensional (1D) heat equation

$$\frac{\partial u(x, t)}{\partial t} = \alpha \frac{\partial^2 u(x, t)}{\partial x^2} \quad (41)$$

where α (or often written in a squared constant) is called the thermal diffusivity of a length- L rod. Its initial condition is

$$u(x, 0) = f(x), 0 < x < L. \quad (42)$$

And we start with the easiest boundary condition, a homogeneous Dirichlet boundary condition

$$u(0, t) = u(L, t) = 0, t > 0. \quad (43)$$

With a 6-point (2 end-points, x_0 and x_5 are fixed by the boundary condition) second order central difference approximation

$$\left. \frac{\partial^2 u(x, t)}{\partial x^2} \right|_{x_j} \approx \frac{u(x_{j+1}) - 2u(x_j) + u(x_{j-1}))}{h^2} \quad (44)$$

we have

$$\frac{d}{dt} \begin{bmatrix} u(x_1, t) \\ u(x_2, t) \\ u(x_3, t) \\ u(x_4, t) \end{bmatrix} = -\frac{\alpha}{h^2} \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} u(x_1, t) \\ u(x_2, t) \\ u(x_3, t) \\ u(x_4, t) \end{bmatrix}, u(x, 0) = f(x) \quad (45)$$

And to decompose the coefficient matrix into Pauli words,

$$T = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix} = 2I_0I_1 - I_0X_1 - \frac{1}{2}(X_0X_1 + Y_0Y_1) \quad (46)$$

And $A = \frac{\alpha}{h^2}T$. Since A is real symmetric,

$$L = \frac{A + A^\dagger}{2} = A \quad (47)$$

$$H = \frac{A - A^\dagger}{2i} = \mathbf{0} \quad (48)$$

1. Suppose we discretize the domain into $N = 2^m$ data points, what is the Pauli representation for T ? Does it still give a Pauli sum representation for L where all Paulis are mutually commuting with each other? If not, how many commuting groups of Paulis are there? The idea is that all Paulis within a group are all commuting so they can be performed in a single Trotter step. Then the total number of Trotter steps needed is only equal to the number of non-commuting groups. “done”
2. One can think about mapping the system itself to another oscillator, and synthesize the required non-Gaussian unitary operator using elementary gate set from ISA? One recent paper developed a new method to synthesize multi-qumode non-Gaussian entangling operation <https://arxiv.org/abs/2512.19582>. “done”
3. Insert a table, summarize the qubit and qumode count, gate count, runtime, as a function of simulation accuracy, time, and diffusion constant. “done”
4. Do the same for the 2D heat equations. “done”

For a grid of N data points, we have a register of $m = \log_2 N$ qubits. We partition the operator A into two sub-operators, $A = A_{\text{even}} + A_{\text{odd}}$, defined by the sets of interacting qubit pairs:

$$A_{\text{even}} = \sum_{j \text{ even}} h_{j, j+1}, \quad A_{\text{odd}} = \sum_{j \text{ odd}} h_{j, j+1}, \quad (49)$$

where the local bond operator

$$h_{j,j+1} = \frac{\alpha}{\hbar^2} \left[I_j I_{j+1} - \frac{1}{2} (X_j X_{j+1} + Y_j Y_{j+1}) \right]. \quad (50)$$

represents the interaction between adjacent spatial grid points.

We now have two sets of local bond operators based on j being even or odd. Operators within each set commute with each other, i.e., $[h_{j,j+1}, h_{k,k+1}] = 0$ for all j, k within each set. This property ensures that the exponential of the sum factorizes exactly: $\exp(-\delta t A_{\text{even}}) = \prod_{j \text{ even}} \exp(-\delta t h_{j,j+1})$. This allows for the simultaneous execution of gates within each layer, reducing the algorithmic depth to a constant factor. For a discrete time step $\delta t = t/n$, the evolution is given by:

$$\mathcal{U}(\delta t) \approx e^{-i\hat{x} \otimes A_{\text{even}} \delta t} e^{-i\hat{x} \otimes A_{\text{odd}} \delta t} = \prod_{j \text{ even}} \exp(-i\delta t \hat{x} \otimes h_{j,j+1}) \prod_{j \text{ odd}} \exp(-i\delta t \hat{x} \otimes h_{j,j+1}) \quad (51)$$

Each Trotter step requires $8(m-1)$ 1-qubit basis rotation gates (H, S), $4(m-1)$ CNOT gates (Two per interaction term (XX and YY)), $m-1$ Conditional Displacement (CD) gates, and 2 circuit layers (one for each even/odd set). The total resource count for p^{th} order Trotterization is given in the first row of table 3.

Alternatively, we can couple the system oscillator to an ancilla oscillator, in which case the ancilla oscillator encodes the spatial basis x in the position operator $\hat{x}$. Then the heat equation becomes

$$\frac{d\mathbf{u}}{dt} = -\alpha \hat{p}^2 \mathbf{u} \quad (52)$$

where $\mathbf{u} := \int u(x, t) |x\rangle dx$. This qumode embedding scheme is already known [8]. However, we can prepare the initial state $\mathbf{u}(t=0)$ using our proposed methodology from 3.1. Then from theorem 2 and ref. [10], we can obtain the evolution:

$$|u(t)\rangle = (\langle \phi |_{\text{osc}} \otimes \mathbb{I}) \hat{U}(|\psi\rangle_{\text{osc}} \otimes |u_0\rangle_{\text{anc}}) \quad (53)$$

where $\hat{U} := R(-\frac{\pi}{2}) e^{-i\alpha t \hat{x} \otimes \hat{x}^2} R(\frac{\pi}{2})$ with $e^{-i\alpha t \hat{x} \otimes \hat{x}^2}$ being the cubic-QND gate. This requires two additional ancilla modes.

3.5.3 Example: Two dimensional Heat Equation

Two dimensional heat equation

$$\frac{\partial u(x, y, t)}{\partial t} = \alpha \left(\frac{\partial^2 u(x, y, t)}{\partial x^2} + \frac{\partial^2 u(x, y, t)}{\partial y^2} \right), u(x, y, 0) = f(x, y) \quad (54)$$

for $0 < x < a$ and $0 < y < b$. With homogeneous Dirichlet boundary conditions

$$u(0, y, t) = u(a, y, t) = 0 \leq y \leq b, t \geq 0 \quad (55)$$

$$u(x, 0, t) = u(x, b, t) = 0 \leq x \leq a, t \geq 0 \quad (56)$$

Again, using central difference method, we have

$$\frac{du_{i,j}(t)}{dt} \approx \frac{u_{i+1,j}(t) - 2u_{i,j}(t) + u_{i-1,j}(t)}{h_x^2} + \frac{u_{i,j+1}(t) - 2u_{i,j}(t) + u_{i,j-1}(t)}{h_y^2} \quad (57)$$

where $u_{i,j}(t) = u(x_i, y_j, t)$.

Let N be the number of interior points (boundary points are determined by boundary condition)

$$N = \lceil a/h_x - 1 \rceil \cdot \lceil b/h_y - 1 \rceil. \quad (58)$$

Suppose there 2 interior points in x -direction and 4 interior points in y -direction, we denote

$$u_1(t) = u(x_1, y_1, t), u_2(t) = u(x_1, y_2, t), \dots \quad (59)$$

$$u_5(t) = u(x_2, y_1, t), u_6(t) = u(x_2, y_2, t), \dots \quad (60)$$

Note that in this scheme, the neighbor point in y direction is ± 1 indices, but neighbor in x direction is $\pm \lceil b/h_y - 1 \rceil$ indices.

For the form of $u_t = -Au(t)$, in our ordering scheme

$$A_{k,k} = \frac{2}{h_x^2} + \frac{2}{h_y^2} \quad (61)$$

And the neighbor in x direction is on the super and sub diagonal

$$A_{k\pm 1,k} = -\frac{1}{h_y^2} \quad (62)$$

And the neighbor in y direction is

$$A_{k,k\pm 4} = -\frac{1}{h_x^2}. \quad (63)$$

An example coefficient matrix is

$$A = \begin{pmatrix} \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 & 0 \\ -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 \\ 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 & -\frac{1}{h_x^2} & 0 \\ 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & 0 & 0 & 0 & -\frac{1}{h_x^2} \\ -\frac{1}{h_x^2} & 0 & 0 & 0 & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 & 0 \\ 0 & -\frac{1}{h_x^2} & 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} & 0 \\ 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} & -\frac{1}{h_y^2} \\ 0 & 0 & 0 & -\frac{1}{h_x^2} & 0 & 0 & -\frac{1}{h_y^2} & \frac{2}{h_x^2} + \frac{2}{h_y^2} \end{pmatrix}.$$

And the decompose is

$$A = \frac{1}{h_y^2} I \otimes T + \frac{2}{h_x^2} I_0 I_1 I_2 - \frac{1}{h_x^2} X_0 I_1 I_2 \quad (64)$$

$$= \frac{1}{h_y^2} \left[2I_0 I_1 I_2 - I_0 I_1 X_2 - \frac{1}{2} (I_0 X_1 X_2 + I_0 Y_1 Y_2) \right] + \frac{2}{h_x^2} I_0 I_1 I_2 - \frac{1}{h_x^2} X_0 I_1 I_2 \quad (65)$$

$$= \left(\frac{2}{h_y^2} + \frac{2}{h_x^2} \right) I_0 I_1 I_2 - \frac{1}{h_y^2} \left(I_0 I_1 X_2 - \frac{1}{2} (I_0 X_1 X_2 + I_0 Y_1 Y_2) \right) - \frac{1}{h_x^2} X_0 I_1 I_2 \quad (66)$$

If we choose arbitrary grid resolutions on both dimensions so that we have $N_x = 2^{m_x}$ points along the x -dimension and $N_y = 2^{m_y}$ points along the y -dimension, then we can map the system to a register of $m = m_x + m_y$ qubits. The 2D Laplacian operator A is then defined by:

$$A = \frac{1}{h_x^2} (T_x \otimes I^{\otimes m_y}) + \frac{1}{h_y^2} (I^{\otimes m_x} \otimes T_y)$$

where T_x and T_y are the 1D discrete Laplacian operators of dimension 2^{m_x} and 2^{m_y} , respectively.

The discretization of A via the central difference method allows for a structured mapping onto the qubit register. The global operator is expressed as a linear combination of Pauli strings:

$$A = \sum_k c_k P_k, \quad P_k \in \{\mathcal{S}_x \cup \mathcal{S}_y\}$$

where the subspaces $\mathcal{S}_x = \{\mathcal{P}_x \otimes I^{\otimes m_y}\}$ and $\mathcal{S}_y = \{I^{\otimes m_x} \otimes \mathcal{P}_y\}$ contain Pauli strings derived from the x and y spatial derivatives, respectively. Because these operators act on disjoint sets of qubits (the x -subspace on the first m_x qubits and the y -subspace on the remaining m_y qubits), any string in $\mathcal{S}_x$ naturally commutes with any string in $\mathcal{S}_y$.

We can then partition $\mathcal{S} = \mathcal{S}_x \cup \mathcal{S}_y$ into three disjoint groups. These groups are defined as follows:

- Group $\mathcal{G}_1$ (Global X-Basis): Comprises the identity and all strings composed strictly of I and X operators from both spatial dimensions. It contains the primary diagonal terms and the neighbor-to-neighbor hopping logic,

$$\mathcal{G}_1 = \{I^{\otimes K}\} \cup \{P \in \mathcal{S}_x \mid P \in \{I, X\}^{\otimes K}\} \cup \{Q \in \mathcal{S}_y \mid Q \in \{I, X\}^{\otimes K}\}$$

- Group $\mathcal{G}_2$ (y-Exchange Mixed Basis): This group includes x -dimension X -type strings to maintain full register utility while measuring the y -subspace in the Y -basis.

$$\mathcal{G}_2 = \{Q \in \mathcal{S}_y \mid Q \notin \{I, X\}^{\otimes K}\} \cup \{P \in \mathcal{S}_x \mid P \in \{I, X\}^{\otimes K}\}$$

- Group $\mathcal{G}_3$ (x-Exchange Mixed Basis): This group includes x -dimension exchange terms while measuring the y -subspace in the X -basis to capture vertical hopping terms.

$$\mathcal{G}_3 = \{P \in \mathcal{S}_x \mid P \notin \{I, X\}^{\otimes K}\} \cup \{Q \in \mathcal{S}_y \mid Q \in \{I, X\}^{\otimes K}\}$$

For example, for an 8×4 grid, we map the coordinates to a $m = 5$ qubit register ($m_x = 3$ for x , $m_y = 2$ for y). Let the first three qubits $\{q_0, q_1, q_2\}$ represent the x -dimension and the last two $\{q_3, q_4\}$ represent the y -dimension. The sets are defined by their functional role and their Pauli alphabet. The Subspace Sets ($\mathcal{S}_x$ and $\mathcal{S}_y$) contain all strings with non-zero coefficients in the global Hamiltonian, categorized by which physical dimension they describe.

- $\mathcal{S}_x$ (x -dimension strings): These act non-trivially on qubits 0, 1, 2 and as Identity on 3, 4.
 - Hopping (X -type): $\{X_0 I_1 I_2 I_3 I_4, I_0 X_1 I_2 I_3 I_4, X_0 X_1 I_2 I_3 I_4, \dots\}$
 - Exchange (Y -type): $\{Y_0 Y_1 I_2 I_3 I_4, I_0 Y_1 Y_2 I_3 I_4, \dots\}$
- $\mathcal{S}_y$ (y -dimension strings): These act non-trivially on qubits 3, 4 and as Identity on 0, 1, 2.
 - Hopping (X -type): $\{I_0 I_1 I_2 X_3 I_4, I_0 I_1 I_2 I_3 X_4, I_0 I_1 I_2 X_3 X_4\}$
 - Exchange (Y -type): $\{I_0 I_1 I_2 Y_3 Y_4\}$

The Commuting Groups ($\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$) are the disjoint partitions of the union $\mathcal{S}_x \cup \mathcal{S}_y$. Within each group, every string commutes. Between groups, they do not.

- $\mathcal{G}_1$: Global X -Basis Group contains the identity and all strings that only use I and X . Example Elements: $\{IIIII, X_0 I_1 I_2 I_3 I_4, I_0 I_1 I_2 X_3 X_4\}$
- $\mathcal{G}_2$: y -Exchange Group contains the Y -type strings for the y -dimension, plus the X -type strings for x . Example Elements: $\{I_0 I_1 I_2 Y_3 Y_4, X_0 X_1 I_2 I_3 I_4\}$
- $\mathcal{G}_3$: x -Exchange Group contains the Y -type strings for the x -dimension, plus the X -type strings for y . Example Elements: $\{Y_0 Y_1 I_2 I_3 I_4, I_0 I_1 I_2 X_3 X_4\}$

For a discrete time step $\delta t = t/n$, the evolution is given by:

$$\mathcal{U}(\delta t) \approx e^{-i\mathcal{G}_3 \delta t} \cdot e^{-i\mathcal{G}_2 \delta t} \cdot e^{-i\mathcal{G}_1 \delta t} \quad (67)$$

Each Trotter step requires $9m$ 1-qubit basis rotation gates (H, S), $6m$ CNOT gates, $\mathcal{O}(m)$ Conditional Displacement (CD) gates [check if it is \$2^m\$ and if there is some pattern in the angle paramters so that we can reduce it to \$m\$, check HHL paper](#), and 3 circuit layers (one for each group). If $N_x \approx N_y$, then $m \approx \log_2 N_x$. The total resource count for p^{th} order Trotterization is given in the second row of table 3.

[do a lit search to see if exponential speedup already exists](#)

The groups are known as maximal commuting sets. For a d -dimensional heat equation, the number of maximal commuting sets is $d + 1 \approx d$. Therefore, we can generalize our resource estimation for a $d - D$ system. The generalized resource requirement is given in the third row of table 3.

Dimension	(qubit, qumode)	1-qubit gate	CNOT gate	CD gate	Circuit Depth
1D	$(\log_2 N, 1)$	$\mathcal{O}\left(\frac{(8t \log_2 N)^{1+1/p}}{\epsilon_t^{1/p}}\right)$	$\mathcal{O}\left(\frac{(4t \log_2 N)^{1+1/p}}{\epsilon_t^{1/p}}\right)$	$\mathcal{O}\left(\frac{(t \log_2 N)^{1+1/p}}{\epsilon_t^{1/p}}\right)$	$\mathcal{O}\left(2t^{1+1/p} \left(\frac{\log_2 N}{\epsilon_t}\right)^{1/p}\right)$
2D	$(2 \log_2 N, 1)$	$\mathcal{O}\left(\frac{(9t \log_2 N)^{1+1/p}}{\epsilon_t^{1/p}}\right)$	$\mathcal{O}\left(\frac{(6t \log_2 N)^{1+1/p}}{\epsilon_t^{1/p}}\right)$	$\mathcal{O}\left(\frac{(t \log_2 N)^{1+1/p}}{\epsilon_t^{1/p}}\right)$	$\mathcal{O}\left(3t^{1+1/p} \left(\frac{\log_2 N}{\epsilon_t}\right)^{1/p}\right)$
dD	$(d \log_2 N, 1)$	$\mathcal{O}\left(\frac{(dt \log_2 N)^{1+1/p}}{\epsilon_t^{1/p}}\right)$	$\mathcal{O}\left(\frac{(dt \log_2 N)^{1+1/p}}{\epsilon_t^{1/p}}\right)$	$\mathcal{O}\left(\frac{(t \log_2 N)^{1+1/p}}{\epsilon_t^{1/p}}\right)$	$\mathcal{O}\left(dt^{1+1/p} \left(\frac{\log_2 N}{\epsilon_t}\right)^{1/p}\right)$

Table 3: Resource cost scaling as a function of runtime t , simulation accuracy ϵ_t , and grid points/dimension N

4 Implementations

5 Experiments

6 Conclusion

7 Future Works

8 Author Contributions

If needed

9 Acknowledgments

10 Competing interests

All authors declare no financial or non-financial competing interests.

11 Data and code availability

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Appendix

Some LCHS basics if needed

Complementary (Homogeneous) Solution $\mathcal{T}e^{-\int_0^t A(s)ds}u_0$

Remark 1. With a quadrature rule, the goal is to find the approximation

$$\int_a^b g(x) \approx \sum_{j=0}^{m-1} w_j g(x_j). \quad (68)$$

For an integral over $[-1, 1]$ with Gaussian quadrature with m nodes, set node x_j to be the root of Legendre polynomial of degree m , $P_m(x)$ with weight $w_j = \frac{2}{(1-x_j^2)((P'_m(x_j))^2)}$. For a longer interval $[a, b]$, the composite rule is applied by dividing the interval $[a, b]$ into K segments, i.e., $[a + kh, a + (k+1)h]$ where $h = \frac{b-a}{K}$ is the step size. Gaussian quadrature is applied on each shorter interval and it is constructed from that on $[-1, 1]$ by a change of variable. That is, the weight $w_j \rightarrow \frac{h}{2}w_j$ and node $x_j \rightarrow \frac{h}{2}x_j + (a + kh) + \frac{1}{2}h$.

Consider the ODE solution over a truncated finite interval $[-K, K]$ using the composite Gaussian quadrature to discretize the interval.

The first step is to define

$$g(k) = \frac{1}{C_\beta(1 - ik)e^{(1+ik)^\beta}}, \quad U(T, k) = \mathcal{T}e^{-i \int_0^T (kL(s) + H(s))ds}. \quad (69)$$

Then,

$$\begin{aligned} \mathcal{T}e^{-\int_0^T A(s)ds} &= \int_{\mathbb{R}} g(k)U(T, k)dk && \text{(by LCHS)} \\ &\approx \int_{-K}^K g(k)U(T, k)dk && \text{(truncation)} \\ &= \sum_{m=-K/h_1}^{K/h_1-1} \int_{mh_1}^{(m+1)h_1} g(k)U(T, k)dk && \text{(divide to shorter intervals)} \\ &\approx \sum_{m=-K/h_1}^{K/h_1-1} \sum_{q=0}^{Q-1} c_{q,m}U(T, k_{q,m}) && \text{(Gaussian quadrature in each interval)} \\ &= \sum_{j=0}^{M-1} c_j U(T, k_j) && \text{(simplified notation)} \end{aligned} \quad (70)$$

where h_1 is the step size in the composite Gaussian quadrature, Q is the number of nodes in each interval $[mh_1, (m+1)h_1]$, $k_{q,m}$ is a Gaussian node, $c_{q,m} = w_q g(k_{q,m})$, w_q is a Gaussian weight (which does not depend on the choice of m), and M is the overall number of the nodes (or unitaries).

Assumption 1. Assume K and h_1 are chosen so that K/h_1 is an integer.

To bound the truncation error by $\epsilon > 0$, it suffices to choose

$$K = \mathcal{O}\left((\log(1/\epsilon))^{1/\beta}\right)$$

To bound the quadrature error by $\epsilon > 0$, it suffices to choose

$$h_1 = \frac{1}{eT \max_t \|L(t)\|}, \quad Q = \left\lceil \frac{1}{\log 4} \log \left(\frac{8}{3C_\beta} \frac{K}{\epsilon} \right) \right\rceil = \mathcal{O}\left(\log \frac{1}{\epsilon}\right), \quad (72)$$

and the overall number of unitaries in the summation formula is

$$M = \frac{2KQ}{h_1} = \mathcal{O}\left(T \max_t \|L(t)\| \left(\log \frac{1}{\epsilon}\right)^{1+1/\beta}\right) \quad (73)$$

For the specific value of K , note that in Eq. (176) in [2], we have

$$\left\| \int_{\mathbb{R}} g(k) U(T, k) dk - \int_{-K}^K g(k) U(T, k) dk \right\| \leq \frac{2^{B+1} B!}{C_{\beta} (\cos(\beta\pi/2))^B} \frac{1}{K} e^{-\frac{1}{2} K^{\beta} \cos(\beta\pi/2)} \leq \epsilon$$

$$\ln \left(\frac{2^{B+1} B!}{C_{\beta} (\cos(\beta\pi/2))^B} \right) + \ln(1/\epsilon) \leq \ln(K) + \frac{1}{2} K^{\beta} \cos(\beta\pi/2) \quad (74)$$

$$2 \cdot \frac{\ln \left(\frac{2^{B+1} B!}{C_{\beta} (\cos(\beta\pi/2))^B} \right) + \ln(1/\epsilon) - \ln(K)}{\cos(\beta\pi/2)} \leq K^{\beta} \quad (75)$$

where $B = \lceil 1/\beta \rceil$

Inhomogeneous Term $\int_0^t \mathcal{T} e^{-\int_s^t A(s') ds'} b(s) ds$

The discretization of the inhomogeneous term is

$$\begin{aligned} & \int_0^T \mathcal{T} e^{-\int_s^T A(s') ds'} b(s) ds \\ &= \int_0^T \int_{\mathbb{R}} \frac{f(k)}{1 - ik} U(T, s, k) b(s) dk ds \quad (\text{by LCHS Eq. (3)}) \\ &\approx \int_0^T \sum_{m_1=-K/h_1}^{K/h_1-1} \sum_{q_1=0}^{Q_1-1} c_{q_1, m_1} U(T, s, k_{q_1, m_1}) b(s) ds \quad (\text{by Eq. (70)}) \\ &\approx \sum_{m_2=0}^{T/h_2-1} \sum_{q_2=0}^{Q_2-1} \sum_{m_1=-K/h_1}^{K/h_1-1} \sum_{l_1=0}^{Q_1-1} c'_{q_2, m_2} c_{q_1, m_1} U(T, s_{q_2, m_2}, k_{q_1, m_1}) |b(s_{q_2, m_2})\rangle \quad (\text{composite Gaussian quadrature}) \quad (76) \end{aligned}$$

$$= \sum_{j'=0}^{M'-1} \sum_{j=0}^{M-1} c'_{j'} c_j U(T, s_{j'}, k_j) |b(s_{j'})\rangle \quad (\text{simplified notation}) \quad (77)$$

where $U(T, s, k) = \mathcal{T} e^{-i \int_s^T (kL(s') + H(s')) ds'}$, h_2 is the step size for the discretization of the time integral, Q_2 is the number of Gaussian quadrature nodes, s_{q_2, m_2} is a node, $c'_{q_2, m_2} = w'_{q_2} \|b(s_{q_2, m_2})\|$, the w'_{q_2} is a Gaussian weight. The $|b\rangle$ implies that it is the normalized vector $b/\|b\|$.

Let

- $\Lambda = \sup_{p \geq 0, t \in [0, T]} \|A^{(p)}\|^{1/(p+1)}$
- $\Xi = \sup_{p \geq 0, t \in [0, T]} \|b^{(p)}\|^{1/(p+1)}$, where (p) refers to the p^{th} -order time derivative
- $\|b\|_{L^1} = \int_0^T \|b(s)\| ds$

To bound the approximation error of Eq. (77) by $\epsilon > 0$, it suffices to choose

$$K = \mathcal{O} \left(\left(\log \left(1 + \frac{\|b\|_{L^1}}{\epsilon} \right) \right)^{1/\beta} \right), \quad h_1 = \frac{1}{eT \max_t \|L(t)\|}, \quad h_2 = \frac{1}{eK(\Lambda + \Xi)}, \quad (78)$$

$$Q_1 = \mathcal{O} \left(\log \left(1 + \frac{\|b\|_{L^1}}{\epsilon} \right) \right), \quad Q_2 = \mathcal{O} \left(\log \left(\frac{T(\Lambda + \Xi)}{\epsilon} \right) \right) \quad (79)$$

and

$$M' = \frac{TQ_2}{h_2} = \mathcal{O} \left(TK(\Lambda + \Xi) \log \left(\frac{T(\Lambda + \Xi)}{\epsilon} \right) \right) = \tilde{\mathcal{O}} \left(T(\Lambda + \Xi) \left(\log \left(1 + \frac{\|b\|_{L^1}}{\epsilon} \right) \right)^{1/\beta} \log(1/\epsilon) \right)$$

As indicated in Eq. (234) and (235) in [2], we have

$$\begin{aligned} \|c\|_1 \frac{eT(\Lambda + \Xi)}{4Q_2} &\leq \frac{\epsilon}{2} \\ \frac{1}{\ln(4)} \ln \left(\|c\|_1 \frac{2eT(\Lambda + \Xi)}{\epsilon} \right) &\leq Q_2, \end{aligned} \quad (80)$$

where $\|c\|_1 = \mathcal{O}(1)$. For computing Q_1 , just use the same formula in Eq. (72) but with the K in Eq. (78).

Summary

With LCHS, Eq. (2) becomes

$$u(T) \approx \sum_{j=0}^{M-1} c_j U(T, k_j) |u_0\rangle + \sum_{j'=0}^{M'-1} \sum_{j=0}^{M-1} c'_{j'} c_j U(T, s_{j'}, k_j) |b(s_{j'})\rangle. \quad (81)$$

where $U(T, s, k) = \mathcal{T}e^{-i \int_s^T (kL(s') + H(s')) ds'}$ and $U(T, k) = U(T, 0, k)$. So, we need oracles circuit to implement the Hamiltonian simulation operator $U(T, s, k)$ and oracles to encode the coefficients c and c' into amplitudes. Then, use LCU to obtain $u(T)$.

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